

Macroeconomic Theory and Policy: Lecture 8 (Ch.6)

Population Growth in the Solow Model

- As observed from the data, population growth is a significant contributor to total GDP growth.
 - ▶ For example, by simple decomposition, population growth in the U.S. accounts for 1.5% of the total 3.5% GDP growth.
- Including population growth in the Solow model is important for studying total GDP growth.
- Assume population grows at a constant rate:

$$L_{t+1} = (1 + \bar{n})L_t$$

- ▶ $\bar{n}$ represents the constant rate of population growth.

Population Growth in the Solow Model

- Recall the capital accumulation equation:

$$\Delta k_{t+1} = \bar{s}y_t - \delta k_t$$

- Dividing both sides by k_t :

$$\frac{\Delta k_{t+1}}{k_t} = \bar{s}\frac{y_t}{k_t} - \delta$$

- The term $\frac{\Delta k_{t+1}}{k_t}$ represents the growth rate of capital per capita.
- Recall $k_t = \frac{K_t}{L_t}$, so $g_k = g_K - g_L$:

$$\frac{\Delta k_{t+1}}{k_t} = \frac{\Delta K_{t+1}}{K_t} - \frac{\Delta L_{t+1}}{L_t} = \frac{\Delta K_{t+1}}{K_t} - \bar{n}$$

Population Growth in the Solow Model

- The capital accumulation function at the aggregate level can be rearranged as:

$$\frac{\Delta K_{t+1}}{K_t} = \bar{s} \frac{Y_t}{K_t} - \delta = \bar{s} \frac{y_t L_t}{k_t L_t} - \delta = \bar{s} \frac{y_t}{k_t} - \delta$$

- Substituting this into the formula from the previous slide:

$$\frac{\Delta k_{t+1}}{k_t} = \bar{s} \frac{y_t}{k_t} - \delta - \bar{n}$$

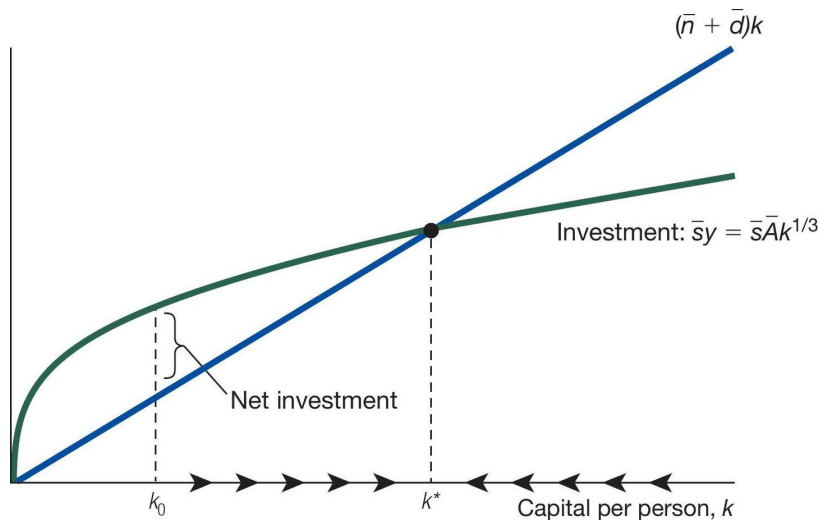
- Multiplying both sides by k_t :

$$\Delta k_{t+1} = \bar{s} y_t - (\delta + \bar{n}) k_t$$

- Compared to the baseline model, investment must not only cover capital depreciation but also provide new capital for the growing population.

Population Growth in the Solow Model

The Solow diagram remains similar to the baseline model, except that the depreciation curve is steeper due to the additional term $\bar{n}$.



Population Growth in the Solow Model

- Adding population growth to the Solow model results in:
 - ▶ Long-run growth in aggregate variables.
 - ▶ No long-run change in GDP per capita, as the economy still converges to its steady state.
 - ★ Population growth effects are already accounted for in the per capita terms. For instance:

$$g_y = g_Y - g_L$$

- ▶ Lower GDP per capita if population growth increases, as more capital investment is needed for new workers, leaving less to raise capital per worker.

What are You Going to Learn (Ch. 6)

- The Romer model of economic growth
- New methods of using existing resources are the key to sustained long-run growth
- The feature of “nonrivalry” makes ideas different from other economic goods
- How the economics of ideas involves increasing returns
- How to combine the Romer and Solow models

The Romer Model

- The Romer model divides the world into
 - ▶ Object
 - ★ Capital and labour from the Solow model
 - ▶ Ideas
 - ★ Ideas used in making objects
 - ★ New ideas are new ways of arranging raw materials in ways that are economically useful
 - ★ e.g., Knowledge in statistic, just-in-time inventory methods, and algebra are also included
- Romer model: long-run economic growth occurs because of the discovery of new ideas

Ideas and Nonrivalry

- Objects are rivalrous when one person's use reduces the usefulness to someone else
 - ▶ GM cannot use Ford's factory to produce cars
- Ideas are nonrivalrous as one person's use does not reduce the usefulness to someone else
 - ▶ Once an idea has been introduced, it can be employed by a number of people without anyone's use being degraded
 - ▶ GM and Ford may learn from each other about building automobile

Ideas and Nonrivalry

- But, ideas maybe *excludability*
 - ▶ There can be legal restrictions on use of a good or idea (e.g., patent, copyright, etc)
 - ▶ But, the person owning the right can distribute the right to other person without the use being degraded
- Research has been conducted on finding the optimal balance between granting inventors exclusive monopoly rights and promoting the dissemination of ideas
- We will exclude this problem for now

Returns to Scale

Recall that:

- Increasing Returns to Scale
 - ▶ Doubling the inputs leads to more than a doubling of outputs
 - ▶ The average production per dollar spent increases as the scale of production grows
- Constant Returns to Scale
 - ▶ Doubling inputs results in exactly doubling the output
 - ▶ The average production per dollar spent remains constant

Returns to Scale

- Return to our standard Codd-Douglas function

- ▶ Suppose the stock of ideas determine the TFP, we include a time subscript on A_t

$$Y_t = F(K_t, L_t, A_t) = A_t K_t^{1/3} L_t^{2/3}$$

- ▶ Here, if A_t does not grow (only the input grow), the function is still constant return to scale

$$F(\gamma K_t, \gamma L_t, A_t) = \gamma A_t K_t^{1/3} L_t^{2/3} = \gamma F(K_t, L_t, A_t)$$

Returns to Scale

- But, with the A_t growing, the function can be increasing return to scale

$$\begin{aligned} F(\gamma K_t, \gamma L_t, \gamma A_t) &= (\gamma A_t)(\gamma K_t)^{1/3}(\gamma L_t)^{2/3} \\ &= \gamma \gamma^{1/3} \gamma^{2/3} A_t K_t^{1/3} L_t^{2/3} = \gamma^2 F(K_t, L_t, A_t) \end{aligned}$$

- If doubling the objects (L,K) is enough to double production, then doubling the objects and the stock of knowledge will more than double production

The Romer Model: Production Function

- Distinction between ideas and objects
- Output requires knowledge and labour
 - ▶ Omitting capital for now
- The production function of the Romer model
 - ▶ Constant returns to scale in objects alone
 - ▶ Increasing returns to scale in objects and ideas

$$Y_t = A_t^\gamma L_{y,t}$$

where $L_{y,t}$ is the number of labour work in production sector in period t

- The parameter γ capture the degree of increasing return to scale – how much idea affects the output
- If the exponent on ideas were less than one, there would still be increasing returns overall (exponents sum to >1)

The Romer Model: New Idea

- New ideas (flow variable) depend on
 - ▶ The number of workers producing ideas, $L_{a,t}$
 - ▶ Worker productivity, $\bar{z}$

$$\Delta A_{t+1} = \bar{z} L_{a,t}$$

- The function is constant return to workers

The Romer Model: Labour and Population

- Population: Workers either producing output or ideas

$$L_{a,t} + L_{y,t} = N_t$$

where N_t is the total population size in period t

- The population is growing at the rate of $\bar{n}$

$$N_{t+1} = (1 + \bar{n})N_t$$

or

$$\frac{\Delta N_{t+1}}{N_t} = \bar{n}$$

The Romer Model: Labour and Population

- Express the endogenous variables in terms of the parameters

$$L_{a,t} = \bar{\ell} N_t$$

where $\bar{\ell}$ is the proportion of labour producing ideas

- Given this, the number of labour work in production are

$$L_{y,t} = (1 - \bar{\ell}) N_t$$

- Meanwhile there are no rivalry for ideas. There is rivalry in labour

Solving the Romer Model

- Five unknown:

$$Y_t, A_t, L_{y,t}, L_{a,t}, N_t$$

- Four parameters:

$$\bar{z}, \bar{L}, \bar{\ell}, \bar{A}_0, \bar{N}_0$$

- Four equations:

$$Y_t = A_t^\gamma L_{y,t}$$

$$\Delta A_{t+1} = \bar{z} L_{a,t}$$

$$L_{y,t} + L_{a,t} = N_t$$

$$L_{a,t} = \bar{\ell} N_t$$

$$N_{t+1} = (1 + \bar{n}) N_t$$

Solving the Romer Model

- Combine the last two equations:

$$L_{y,t} = (1 - \bar{\ell})N_t \implies \frac{L_{y,t}}{N_t} = (1 - \bar{\ell})$$

- Output per person depends on the stock of knowledge

$$y_t = \frac{Y_t}{N_t} = A_t^\gamma \frac{L_{y,t}}{N_t} = A_t^\gamma (1 - \bar{\ell})$$

- ▶ Romer model: Output per person depends on the total stock of knowledge
- ▶ Solow model: Output per person depends on capital per person

Solving the Romer Model

- The growth rate of knowledge is

$$g_{A,t} = \frac{\Delta A_{t+1}}{A_t} = \frac{\bar{z}L_{a,t}}{A_t}$$

- ▶ The growth rate of idea is proportional to the number of researchers in the economy, the research productivity and existing stock of idea
- Solving for the stock of knowledge:

$$A_t = \frac{1}{g_{A,t}} \bar{z}L_{a,t}$$

Solving the Romer Model: Balanced Growth Path

- The growth of knowledge, g_A , is positive given that $\bar{z}$ and $L_{a,t}$ cannot be zero
 - ▶ This implies continuous growth in knowledge
- In the long run, the model converges to a specific growth rate, known as the *Balanced Growth Path* (BGP)
 - ▶ This occurs when all variables grow at a constant rate
 - ▶ These variables do not necessarily have to grow at the same rate
- The concept is similar to a steady state, but applies to growth rates

Solving the Romer Model: Balanced Growth Path

- Consider the knowledge stock formula: $A_t = \frac{1}{g_{A,t}} \bar{z} L_{a,t}$.
- In the BGP, $g_{A,t} = g_A^*$, making $\frac{1}{g_{A,t}}$ and $\bar{z}$ constants
- Using the growth rate formula and noting that the growth rate of a constant is zero:

$$g_A = g_{L_a} = \bar{n}$$

- ▶ The equality $g_{L_a} = \bar{n}$ holds because a fixed proportion ℓ of the population works in the research sector
- ▶ The growth of L_a follows the population growth rate $\bar{n}$

Solving the Romer Model: Output per Capita

- Substituting the stock of knowledge and $g_A = \bar{n}$ into the output per capita equation:

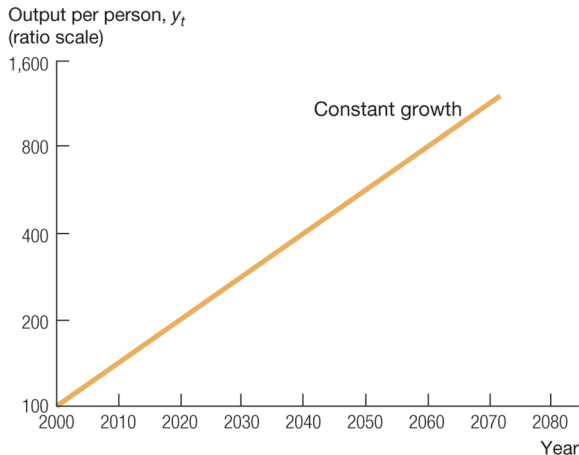
$$\begin{aligned}y_t^* &= A_t^\gamma (1 - \bar{\ell}) \\&= \left(\frac{1}{g_{A,t}} \bar{z} L_{a,t} \right)^\gamma (1 - \bar{\ell}) \\&= \left(\frac{1}{\bar{n}} \right)^\gamma \bar{z}^\gamma (L_{a,t})^\gamma (1 - \bar{\ell})\end{aligned}$$

- All terms are constant except for $L_{a,t}$

$$g_y^* = \gamma \bar{n}$$

Output per Person

- Output per person grows at a constant rate and so, it appears as a straight line on a ratio scale



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Why Is There Growth in the Romer Model

The Romer model produces long-run growth

- Does not have diminishing returns to ideas because they are nonrivalrous
- Together, labour and ideas have increasing returns
- Returns to ideas are unrestricted – R&D leads to sustained growth of income over time
- As the economy accumulates new ideas, the returns to knowledge does not fall (i.e., no depreciation)
 - ▶ In the Solow model, capital has diminishing returns, capital and income stop growing eventually

Case Study: A Model of World Knowledge

- The United States has more researchers than Luxembourg has people
- According to the Romer's model, the growth rate of GDP per capita in the model is tied to the number of researchers
 - ▶ Therefore, the model would predict that the United States should grow faster than Luxembourg
- Between 1960 and 2017,
 - ▶ U.S. growth rate of per capita GDP: 2.0 percent per year
 - ▶ Luxembourg growth rate of per capita GDP: 2.7 percent per year
- All countries can benefit from all ideas, no matter where the ideas were discovered
 - ▶ e.g., International trade, multinational corporations, licensing agreements, international patent filings, the migration of students and workers, and the open flow of information

Experiment: An Increase in Population

- What happen if a country suddenly one time increase its population? In BGP, ...

$$y_t^* = \left(\frac{1}{\bar{n}}\right)^\gamma \bar{z}^\gamma (1 - \bar{\ell}) (L_{a,t})^\gamma = \left(\frac{1}{\bar{n}}\right)^\gamma \bar{z}^\gamma (1 - \bar{\ell}) (\ell N_t)^\gamma$$
$$g_y^* = \gamma \bar{n}$$

- Changes in the population can be interpret as an exogenous increase in N_t
 - ▶ A larger population means there are more researchers (i.e., L_a increases) lead to more idea being created, so the higher per capita output
- But, there is no change to g_y
- Therefore, an increase in population permanently raises per capita output, but after the rise, the growth is the same as before

Experiment: An Increase in Population

- What happens during the period before reaching BGP?
- The economy will follow the transitional dynamics as discussed in the previous chapter. Why does this occur in the Romer model?
- In the initial period, y_t remains unchanged.
 - ▶ Recall $Y_t = A_t^\gamma L_{y,t}$. Y increases due to the rise in workers in the production sector (i.e., $L_{y,t} = (1 - \ell)N_t$)
 - ▶ In our specification, this cancels out the effect of the increase in N , resulting in no immediate change in y_t (i.e., $y_t = A_t^\gamma (1 - \ell)$)

Experiment: An Increase in Population

- Some of the new population will enter the research sector, accelerating the rate of idea creation (i.e., $\Delta A_{t+1} = \bar{z}L_{a,t} = \bar{z}\ell N_t$).
- As A_{t+1} increases, y_{t+1} will grow at a faster rate than the increase in N_{t+1} would suggest, leading to a faster rise in y
- This accelerated growth will eventually slow as the $g_A = \frac{\bar{z}L_{a,t}}{A_t}$ returns to the rate of $\bar{n}$ on the BGP

Growth Effects versus Level Effects

- Growth effects: changes to the rate of growth of per capita output
 - ▶ Changes that affect g 's
- Level effects: Changes in the level of per capita GDP
 - ▶ Changes affect y , Y , and so forth

Next Lecture

- Next lecture will finish Ch.6